

C Thomas, PhD

(they/them)

[Website](#) [LinkedIn](#) [GitHub](#) [Scholar](#) [email](#) [SF Bay Area](#)

Professional Summary

- **AI R&D Leader bridging frontier science and real-world constraints.** Open to Head/Director of Applied AI roles, or high-impact research leadership focused on frontier model training, evaluation, and interpretability.
- **Expert in bridging research and engineering cultures,** aligning technical and executive teams to achieve high-stakes milestones across academia, startups, and enterprise. Consistently identify gaps and build new teams to solve them.
- **Built and lead Apple R&D team** developing novel, robust models for manufacturing product quality. Direct a core ML toolkit used by 20+ MLEs and 5+ teams to deploy breakthroughs across hundreds of global factory lines.
- **Cut operational costs by 60% and accelerated model deployment from weeks to hours** by bringing engineering and science into one team, grounding work in factory realities, and driving end-user adoption.
- **Actively advancing fundamental research** on LLM mechanics and memory-augmented architectures (ICLR, 2026).

Skills & Tools

AI/ML Methods:	LLM Pretraining, Computer Vision & VLMs, Domain Adaptation, LoRA Fine-Tuning, Synthetic Defect Generation, Evals
Tools:	Python, PyTorch (DDP), Hugging Face, LLM Foundry, Detectron2, OpenMMLab
Engineering & Scale:	Systems Engineering, Automated Evaluation & Stress-Testing, MLOps Infrastructure
R&D Leadership:	Research-to-Production Translation, Cross-Functional Strategy, Team Vision & Engagement, Academic Partnerships, Mentorship

Experience

Apple Cupertino, CA

Applied Research Manager, Machine Learning

2022 – current

- **Built the applied ML research function,** directing the development of core training libraries that enable 20+ MLEs and 5+ cross-functional teams to leverage research breakthroughs across hundreds of global deployments.
- **Scaled organizational research capacity** by unifying ad-hoc university collaborations, guiding 3 internal PIs launching external initiatives, and leading an 80+ person summit to secure broad alignment and research buy-in.
- **Fostered career growth over 5+ years of ML leadership,** managing 4-6 direct reports and mentoring 5+ students; built individualized development plans at the intersection of their goals and the team's mission.
- **Accelerated image model updates from weeks to hours** by partnering with Ops and Hardware to identify bottlenecks; deployed a LoRA defect injection platform that reduced synthetic data needs from 10,000 to 5 images.
- **Reduced factory annotation costs by up to 60%** by integrating self-supervised frameworks into production systems to solve cross-generation device adaptation challenges.
- **Engineered the evaluation pipeline for a novel memory-augmented LLM** (ICLR 2026); authored a patch to LLM Foundry, built the memory-retrieval routing logic, and executed end-to-end evaluations comparing baseline and augmented architectures.

Machine Learning Engineer

2020 – 2022

- **Reduced urgent model deployment time by 80%** by implementing MLOps infrastructure; bridged historical divides between ML, Data Engineering, and Software teams to build a unified data management platform.
- **Trained production computer vision models** for deployment on factory lines, building monitoring systems to detect model drift and core vulnerabilities prior to customer impact.
- **Established engineering standards** and mentorship to junior ICs, interns and student collaborators in collaborative software development and rigorous experimental data interpretation.

Tua Fertility Oakland, CA

Founder & CEO

2019 - 2020

- **Designed, patented and prototyped a medical device,** executing end-to-end mechanical design (CAD/SolidWorks) and rapid iterations to proof of concept as a resident at the Autodesk Technology Center
- **Drove technical vision and business strategy,** pitching to investors and translating complex engineering roadmaps into compelling narratives.

Rigetti Computing Berkeley, CA

Senior Research Engineer & Tech Lead

2017 - 2019

- **Drove multiple quantum hardware breakthroughs** by aligning fabrication, cryogenic, and electrical teams and subsystems to stabilize and optimize complex, non-linear computing architectures.
- **Built robust R&D software** to calibrate and run quantum gate operations, analyzing large datasets and distilling results into shared tooling for research teams.
- **Influenced executive technical roadmaps**, regularly translating complex engineering progress into actionable milestones that shifted company timelines and targets.

University of California, Berkeley Berkeley, CA

Doctoral Research Fellow

2010 - 2017

- **Awarded the highly competitive DOE SCGF (accepted) and NSF (declined) Fellowships.**
- **Designed and built ultra-high vacuum and laser systems** to trap individual atoms in lattices of laser light, combining optics, analog electronics, and hardware with analytical mean-field modeling to explain observed quantum phase transition ([Phys. Rev. Lett., Editor's Suggestion](#)).
- **Resolved a contested phenomenon in quantum literature** by developing a predictive white-box model that provided a clear explanation for unexpected asymmetries in 2D absorption image data ([Phys. Rev. A](#)).

Selected Publications

Pretraining with hierarchical memories: separating long-tail and common knowledge.

H. Pouransari, D. Grangier, [C. Thomas](#), M. Kirchhof, O. Tuzel. *ICLR 2026*. [[arXiv](#)]

Rethinking Semi-Supervised Domain Adaptation for Semantic Segmentation in the Era of Foundation Models. J. Kurien*, B. Balaji*, K.H. Lai, P. G. Vela, [C. Thomas](#), A. Wong, S. Rambhatla. *CVPR 2025 (FGVC Workshop)*.

LangDA: Language-guided Domain Adaptive Semantic Segmentation. C. Liu, S. Hossain, [C. Thomas](#), K.H. Lai, R. Vemulapalli, S. Rambhatla, A. Wong. *NeurIPS 2024 (AFM Workshop)*. [[NeurIPS](#)] [[arxiv](#)]

Synth4Seg: Learning Defect Data Synthesis for Defect Segmentation using Bi-level Optimization. S. Mou, R. Vemulapalli, S. Li, Y. Liu, [C. Thomas](#), M. Cao, H. Bai, O. Tuzel, P. Huang, J. Shan, J. Shi. *IEEE T-ASE 2024*. [[arXiv](#)]

Service & Leadership

Academic Conference Leadership

Co-organizer, 4th Workshop on Vision-based Industrial Inspection , CVPR	2026
Invited Speaker, "Applied Research in Industrial AI", 3rd Workshop , ICCV	2025
Co-organizer & Panelist, Data-efficient defect segmentation, 2nd Workshop , ECCV	2024

Career Mentorship

Committee Member, Employee Career Development & Rotations, Apple Operations	2023 – Current
Career Mentor, 1:1 guidance for recent graduates entering the workforce	2023 – Current

Community

Impact Bay Area is a nonprofit in the SF Bay Area that trains women and girls to set and hold boundaries with verbal and physical self-defense skills.

Board Secretary	2023 – 2025
Assistant Instructor	2017 – 2025

Education

PhD	University of California at Berkeley, Atomic and Optical Physics Quantum simulation of triangular, honeycomb and kagome crystal structures using ultracold atoms in lattices of laser light	2017
MA	University of California at Berkeley, Physics	2013
BA	Boston University, Physics and Math	2010